1 (a) Find the reciprocal of 20

Reciprocal = 1 ÷ 20 = 0.05

1 mark
A reciprocal multiplies the original number to make 1.

0.05
(1)

(b) Work out the value of $\dfrac{3^{220}}{3^{190} \times 3^{28}}$

$3^{190} \times 3^{28} = 3^{218}$

1 mark
Add indices when multiplying same bases.

$\dfrac{3^{220}}{3^{218}} = 3^2 = 9$

1 mark
Subtract denominator index: 220 − 218 = 2.

9
(2)

Asma was asked to write $(5^{60})^3$ as a power of 5

Here is her working.

$(5^{60})^3 = 5^{60+3}$

$= 5^{63}$

(c) Is Asma's working correct?
You must give a reason for your answer.

No. A power of a power multiplies the indices.

$(5^{60})^3 = 5^{180}$, because 60 × 3 = 180.

1 mark
Explain the index rule and correct the error.

(1)

(Total for Question 1 is 4 marks)

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